HEAVY METAL TESTED & CERTIFIED STANDARDS

Heavy Metal Tested & Certified

Team Primer

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Team Primer

This is the short version. It is written for people joining or supporting the Heavy Metal Tested and Certified program who do not yet know how it works, and it aims to get everyone onto the same page in about fifteen minutes of reading.

It is deliberately brief. It tells you what the program is, walks the full path a brand travels from first inquiry to a certified mark, and puts the analytes, statuses, numbers, and terms you will hear most often on a single page. It is not the rulebook. Every rule here is stated in summary; the binding detail, the decision rules, the testing schedule, the enforcement and appeals machinery, and how the limits themselves are set, lives in the full Program Manual, which this primer points you to at the end.

This primer is for internal use. It contains no brand-identifying information and none should be added to it. Read it first; keep the full manual on hand for when you need the exact rule.

Orientation

This section orients a reader who is new to the program. It states in plain terms what the program is, walks the full path a brand travels from first inquiry to a certified mark, and puts the numbers and terms you will meet most often on a single page. It is a way in, not a substitute for the parts that follow: every rule summarized here is specified in full in Part 2 through Part 4.

Start here: what the program is

The Heavy Metal Tested and Certified program does one narrow thing and does it in the open. It publishes a numeric limit for each heavy metal in each kind of product, and then it certifies individual products against those published numbers under continuous laboratory testing. The limit is published before anything is certified against it, so a brand can read the exact threshold it will be held to before it applies or spends a dollar on testing.

Three choices make the program different from the testing a team member may already have in mind. First, the number comes first: the program does not test products and then decide what counts as a pass, it publishes the pass line in advance and holds everyone to it. Second, it is continuous, not a snapshot: a product is certified across multiple production lots and then kept under surveillance for as long as it carries the mark, so the mark describes an ongoing state rather than one lucky test. Third, it tests eight metals as ten analytes, where most food certifications test four, because a limit is only as protective as the metals it actually looks for.

The program is careful about what the mark claims. It certifies that a product was tested at an accredited laboratory, at or below a published number, on a stated analyte panel, in the form it is actually sold in. It

does not claim the product is safe, and a brand may not use it to claim "heavy metal free." That restraint is deliberate and it is what keeps the mark defensible.

One structural point matters from the first day. The program runs on two separate properties. The certification program, its limits, and its governance live at heavymetalcertified.com and are run by the Institute of Contaminant Standards. The scientific evidence the limits draw on lives separately at the Heavy Metal Index, heavymetalindex.com, a public library of the peer-reviewed and regulatory literature. The program reads from the Index; the Index does not endorse the program or any brand. Keeping those two apart is what lets the program cite an independent evidence base rather than its own marketing, and holding that separation is part of everyone's job here.

The certification journey

Here is the whole path a brand travels, first contact to a certified mark. Each phase is specified in full in the part named beside it.

Before any of it, a mutual confidentiality undertaking, pre-signed by the program, is in force from first contact. The brand's own counter-signature is optional and gates nothing. This is the only binding instrument in place before certification, and it is what lets a brand test and build a baseline without creating any public record.

Phase 0. Readiness (free and private). A brand reads the published limit for its subcategory and tests privately against it, entering its own prior laboratory results to see exactly where it stands. Readiness is free and nonpublic: no mark, no fee, and no public record that the brand looked. It exists so that testing is a diagnostic a brand uses to build a plan, not a trap that penalizes it for checking.

Phase 1. The three-lot baseline. A product is certified only after a baseline of three production lots, and a single passing lot certifies nothing. A brand that already runs an accredited testing program may credit up to two of the three lots with qualifying prior results, analyte by analyte. The third, completing lot is one the program runs itself: it purchases a unit of the finished product at retail, ships it under chain of custody to an accredited laboratory, and runs the panel. Buying the product independently, rather than trusting a brand-supplied sample, is what makes the

completing lot a defensible anchor for the baseline, and it is the point at which program cost first enters. Part 2.3 and Part 4.3 specify this.

Phase 2. Application and pathway. A qualifying three-lot baseline unlocks the certification application. On application, the program sets the product's ongoing lot-testing schedule and determines which analytes are assayed on each future lot, and the readiness engine routes the product by its baseline result: a conforming product (Status A or B) proceeds toward certification, and a product not yet meeting the standard enters the confidential readiness track to correct the cause first. A conforming brand also chooses here whether its certification will be public-facing or private. Part 2.4 covers the statuses; Part 2.5 covers the pathways and that choice.

Phase 3. Approval and agreement. After the program's review and approval, the binding Certification Agreement that licenses the mark is drawn up and executed. Only at this point is the product certified, and only a brand that chose the public-facing path is listed publicly. The agreement is the late instrument, not the entry ticket.

Phase 4. The mark and surveillance. A certified product may carry the mark under the mark-usage rules and enters continuous surveillance. How often it is tested depends on its status and risk tier, and any change to ingredients, water, equipment, packaging, formulation, or process triggers an unscheduled reflex test. Part 2.6 covers the mark; Part 4 covers the schedule.

Phase 5. When a lot falls short. If a certified product later fails a lot, the brand opens a corrective action, fixes the root cause, and re-tests, on a clock that tightens with severity. The brand chooses how the finding is disclosed: confidentially by default, or, electively, published on its own brand page as a documented catch-and-fix. Neither route avoids the correction, and neither shields a released lot that is above its limit. A brand that conceals a known exceedance loses its certificate. Part 2.5 and Part 2.9 specify this.

Phase 6. The ratchet (over time). Published limits tighten as the certified market improves, on a capped, pre-announced schedule, and never loosen. A brand can see the next move coming through a published forward forecast and plan for it. Part 2.11 specifies the mechanism.

The path in one line: a confidentiality undertaking is already in force, the brand tests privately for free, builds a three-lot baseline (two credited prior lots plus one the program buys and tests), applies once the baseline qualifies, is routed to certification or to readiness, chooses public-facing or private, and, after approval, executes the agreement that licenses the mark, then stays under surveillance and improves as the limits ratchet down.

Phase	What happens	The gate	Specified in
0. Readiness	Brand tests privately against the published limit, for free	None; nonpublic, no mark, no fee	Part 2.3, 2.5
1. Baseline	Two credited prior lots plus one completing lot the program buys and tests	Three qualifying lots	Part 2.3, 4.3
2. Application and pathway	Baseline unlocks the application; schedule and analytes set; route and public-or-private chosen	Qualifying baseline	Part 2.4, 2.5
3. Approval and agreement	Program approves; Certification Agreement executed; product certified	Program approval	Part 2.5
4. Mark and surveillance	Mark on a certified product; testing by status and risk tier	Mark-usage rules	Part 2.6, Part 4
5. Corrective action	Fix, re-test, disclose confidentially or publicly	No concealment	Part 2.5, 2.9
6. Ratchet	Limits tighten on a capped, forecast schedule	Two-cycle trigger, 20 percent cap	Part 2.11

Program at a glance

The panel. Eight priority metals, tested as ten analytes. Two of the eight are speciated to the toxic form; methylmercury is a conditional eleventh, speciated by reflex and bounded by the total-mercury limit.

Metal	Analyte tested	Tier	Transitional variance
Lead (Pb)	Total	1	None
Cadmium (Cd)	Total	1	None
Arsenic (As)	Inorganic (iAs), speciated	1	None
Mercury (Hg)	Total; methylmercury on trigger	1	None
Nickel (Ni)	Total	2	To 150 percent
Tin (Sn)	Total	2	To 150 percent
Aluminum (Al)	Total	2	To 150 percent
Chromium (Cr)	Total; hexavalent on trigger	2	To 150 percent, Tier 1 when speciated

Tier 1 metals have no safe threshold and admit no exceedance at any margin. Tier 2 metals are controllable and admit a transitional exceedance to 150 percent under conditions.

The five statuses. A status attaches to a product, not a brand, and is visible on the certificate.

Status	Meaning	Mark
A	Certified, full compliance	Usable
B	Certified, transitional (a Tier 2 metal to 150 percent, with a correction plan)	Usable, disclosed
C	Probation (a Tier 1 exceedance, or 150 to 200 percent of limit)	Suspended
D	Suspension (above 200 percent, or a hard-stop such as a regulatory or data-integrity failure)	Prohibited

Status	Meaning	Mark
E	Revocation (concealment or repeated failure)	Prohibited

The numbers you will meet most. A limit is met at 100 percent of the published value or below (percent-of-limit, PoL). Transitional certification runs to 150 percent, on Tier 2 metals only. Above 200 percent is suspension. Entry requires three lots. Corrective action opens within 10 business days at Status B, 5 at Status C, and 2 at Status D. Speciation is triggered when a total metal reaches 50 percent of its limit. The ratchet moves at most 20 percent per event, with a 12-to-18-month transition, and only after two consecutive annual reviews confirm the market has moved.

The terms you will meet most. An *action level* is the limit that triggers intervention; it is not a *safety threshold*, and the difference is load-bearing. *ALARA* is the obligation to keep reducing even while compliant. *Native basis* means a limit is stated in the form the product is sold in, powder as powder, ready-to-feed as consumed. *PoL* and *MaxPoL* are a result as a percentage of its limit, and the highest such across a product's analytes, which sets the status. *Reflex testing* is an unscheduled test triggered by a change or a warning result. *CAPA* is the corrective-and-preventive-action process. *Chain of custody* is the unbroken documented handling of a sample; a break invalidates the result. The *ratchet* is the mechanism that tightens limits over time. The full glossary is in Part 5.5.

Where to Go for More

This primer covers the shape of the program. When you need the exact rule, the full Program Manual is the reference, and it is organized so you can go straight to what you need.

- **How a product gets certified, in binding detail.** Program Manual Part 2: the principles, the analyte panel and tiers, the three-lot baseline, the five statuses, public-facing and private certification, the readiness and corrective-action tracks, the mark rules, and the decision rules.
- **The testing and surveillance schedule.** Part 4: risk tiers, how often each status is tested, the reflex triggers, speciation, sampling, and chain of custody.
- **How the limits themselves are set.** Part 3.1: the government-maximum default and the disclosed occurrence calibration beneath it, the publication gates, and the rationale tags.
- **Why the program holds up under challenge.** Part 5: the methodology defense, the program-level adversarial read, the recognition landscape, and the governance structure.
- **How the program is organized and run.** The internal edition of the manual carries Part 6, Program Operations: the entity structure, the firewall, the standards pipeline, the roles, and the governance workflow.

If you read one thing after this primer, read Part 2. It is the operational constitution, and it answers almost every “what happens if” question a brand or a colleague will ask you.